Nate Haris

43 Lincoln St, Lexington, MA | +1-978-808-5714 | nh428@cornell.edu | nh428.github.io

Education

Cornell University Graduation June 2028
Department of Electrical Engineering | GPA: 3.6 Ithaca, NY

– *Relevant Coursework:* Circuit Analysis, Computer Architecture, Embedded OS Design, Computer Systems Programming, Microelectronics Design

Experience

Electrical Engineer – Howe & Howe Technologies May 2026 – Present
Waterboro, ME

* Co-led proprietary vehicle harness ACES project, created 100+ wiring schematics. Developed custom VBA scripts to automate revision process, increasing efficiency by 33%.
* Created test plans for actuators, motors, and battery systems, verifying functionality under realistic vehicle conditions.

Senior Audio and Visual Technician – Cornell A&S AV Nov. 2024 – Present
Ithaca, NY

* Responsible for maintaining over 200+ integrated AV systems across campus. Researched, implemented, and tested up new technologies, such as touch panels and video streaming services.
* Led onboarding process of 6 new employees, developed flowcharts to aide with common problems.

Student Manager – Cornell Engineering Learning Lab Oct. 2024 – Present
Ithaca, NY

* Overseen operations of lab space for 36 project teams of 1,800+ students with $100,000s of annual funding.
* Taught safe operation practices of heavy machinery, CNCs, lathes, milling machines, welding rigs, and high voltage systems.

Projects

Projects are covered in depth in my personal portfolio: nh428.github.io

Custom Dashboard PCB

* Developed, brought up, and tested custom dashboard PCB and driver modules for Cornell's FSAE race car, learning about PCB design principals. Designed custom software for high performance vehicle diagnostics.

Pipelined Multi-Core Processor

* Designed 4 core pipelined processor in Verilog to execute a subset of RISC-V assembly commands, optimized for sorting algorithms with layered branch target buffers, parallel cache access, and fair ring network arbitration.

Autonomous LiDAR Guide Bot

* Designed an autonomous robotic guide assistant for visually impaired. Used ROS2 LiDAR odometry to map areas then communicate with embedded step tracker module to guide a person around obstacles.
* Developed 3 custom pathfinding algorithms for various environments. Created control module to dynamically select algorithm in real time by accuracy. Currently working on custom FPGA LiDAR and wheel encoder processing module to improving accuracy and decrease latency.

Activities

- Volunteer Firefighter at local department, Cayuga Heights
- Cornell Club-A Team Ultimate Frisbee Player
- CKB Residence hall Vice-President
- Cybersecurity Club member

Skills

Firmware Skills: Verilog, C, C++, Java, Assembly, Python, Html, MATLAB
Technical Skills: PCB design in Altium, Circuit analysis with LTSPICE, FPGA development in Quartus/Ice Studio, AutoCAD, Software development, Cybersecurity vulnerability analysis